VIII a) What are the practical applications of epicyclic gear trains. (5)

b) A pinion 'A' has 15 teeth and is rigidly fixed to a motor shaft. The wheel B has a 20 teeth and gears with A and also with a fixed annular wheel D. The pinion C has 15 teeth and is fixed to the wheel B and gears with annular wheel E which is keyed to the machine shaft.. B and C can rotate together on a pin carried by an arm which rotates about the shaft on which A is fixed. If the motor runs at 1000 rpm, find the speed of the machine. (15)

IX a) Derive an expression for the ratio of belt tension on the tight side and slack side in terms of coefficient of friction, angle of contact and groove angle for V-belt. (8)

b) A V-belt having an area of cross-section 535.5 mm^2 has a groove angle of 30° and an angle of lap of 160°, $\mu = 0.1$. The mass of the rope is 1.45 Kg/m run. The maximum safe stress is 850 N/cm^2. Calculate the power that can be transmitted at 25 m/s. (12)

OR

X a) Derive an expression for the torque transmitted by a single plate clutch assuming uniform pressure theory. (8)

b) A multi-plate clutch is required to transmit 90 kw at 3000 rpm. The plates are alternatively of steel and phosphor bronze and they run in oil. The coefficient of friction is 0.08. The internal radius of the friction surfaces is 0.8 times the radius of the external surfaces. The axial pressure is limited to 20 N/cm^2. If the maximum diameter of the frictional surfaces is not to exceed 250 mm, determine the number of plates required. (12)

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Code No.BTS 118(E)

B.Tech. Degree V Semester Examination in Mechanical Engineering (CAD/CAM), March 1999

ME 501 MECHANICS OF MACHINERY

Time: 3 Hours *Max.Marks: 100*

I a) What is meant by Coriois component of acceleration. (5)

b) The driving crank AB of the quick-return mechanism, as shown in figure 1, revolves at a uniform speed of 200 rpm clockwise. Find the velocity and acceleration of the tool box 'R', in the position shown, when the crank makes an angle of 60° with the vertical line of centres PA. (15)

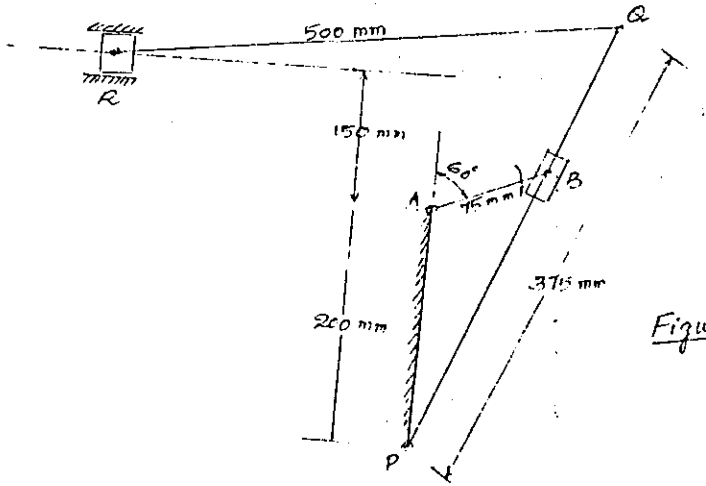

Figure - 1

OR

II a) State and prove the Kennedy's Theorem of three centers.(8)

b) Link 2 of the mechanism shown in figure 2 has an angular velocity of 56 rad/s counter clockwise. Find the velocity at point C. The dimension of various links are as follows: $AO_2 = CB = 150$ mm, $BA = BO_4 = 250$mm, $O_4 O_4 = 100$ mm, $CA = 300$ mm. (12)

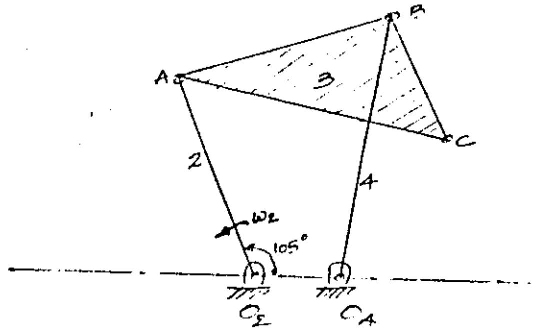

Figure. 2

III a) Discuss the objectives of kinematic synthesis. (5)

b) Lay out a four bar mechanism such that

$\theta_{12} = 40^\circ$, $\theta_{23} = 35^\circ$ for input crank and

$\phi_{12} = 45^\circ$ *and* $\phi_{23} = 65^\circ$ for the output crank. The input crank rotates in counter clockwise direction while the output rotates in a clockwise direction. (15)

OR

IV a) Obtain the Freudenstein's equation for a four bar mechanism. (5)

b) Synthesize a four bar linkage using Freudensteins's equation to generate the function $y = x^{1.8}$ for the interval $1 \le x \le 5$. The input crank is to start from $\theta s = 30^\circ$ and is to have a range of 90°. The output follower is to start at $\phi_3 = 0^\circ$ and is to have a range of 90°. Take three accuracy points at x = 1, 3 and 5. (15)

Contd...3

V Construct the displacement diagram and the cam profile for a plate cam with an oscillating radial flat-face follower which rises through 30° with cycloidal motion in 150° of counterclockwise cam rotation, then dwells for 30°, returns with cycloidal motion in 120°, and dwells for 60°. Determine the necessary length for the follower face, allowing 5mm clearance at each end. The prime circle radius is 30mm. The follower pivot is 120mm to the right. (20)

OR

VI a) Explain the term: "equivalent link mechanism in cam". (5)

b) The exhaust valve of a gas engine working on Otto cycle has a lift of 20mm and is operated by a cam giving constant acceleration and retardation. The valve opens 55° before ODC and closes 15° after the IDC of the crank. The minimum radius of the cam is 3.75 cm. Period for opening of the valve is the same as the period for closing of the valve.
Draw the cam profile and determine the acceleration and retardation and the maximum velocity during the valve opening if the engine runs at 500 rpm. (15)

VII a) Derive an expression for the length of path of contact for two profile gears in mesh. (8)

b) The following data relate to 2 involute gears in mesh. Number of teeth on wheel = 60, on gear = 40. Pressure angle = 20°. module = 8mm. The addendum on each gear is such that the path of approach and recess are 40% of maximum possible. Find the addendum of each gear and length of arc of contact. (12)

Contd...4

OR.